ABSTRACT OF THE DISCLOSURE

A system and method for an ultra-high-bandwidth memory interface that replaces conventional electrical pin-based connectors with a massively parallel optical interconnect. The system comprises a memory module equipped with a high-density array of photon sensors and a corresponding motherboard memory slot equipped with a high-density array of photon emitters. Data is transferred between the processor and the memory module as a sequence of two-dimensional frames of light, where each pixel in the frame acts as an individual data channel. This architecture provides a memory bus with a theoretical bandwidth in the petabit-per-second range, effectively eliminating the memory wall bottleneck that limits the performance of modern processors in data-intensive applications.